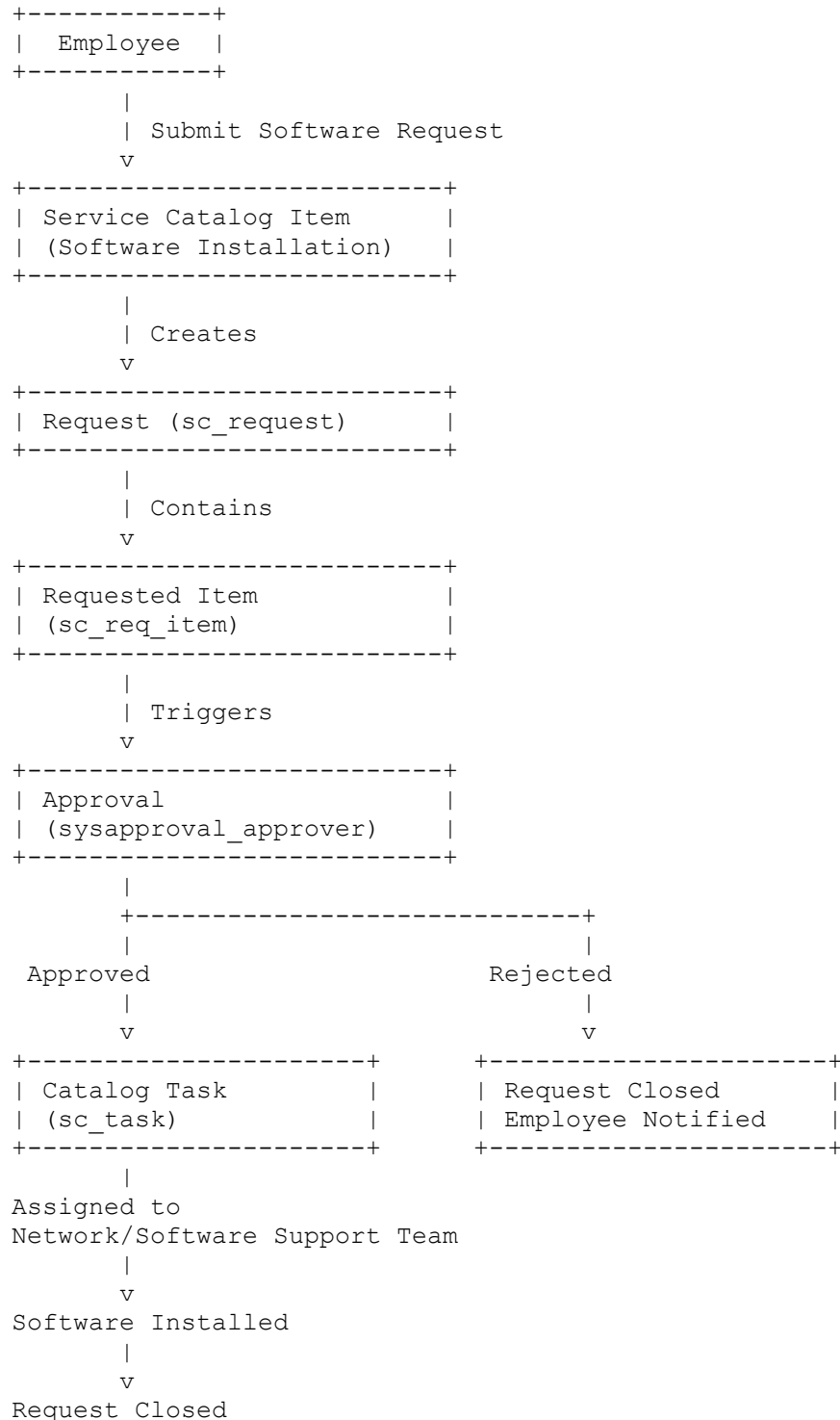


Data Flow Diagrams and User Stories

Project: Employee Software Installation Request (ServiceNow)

1. High-Level Data Flow Diagram (Level 0)



If the software cannot be installed because a license is unavailable, the request is placed ****On Hold****, and a ****Business Rule**** automatically creates an ****Incident (incident)**** assigned to the Software Support team for further action.

2. Data Flow Description

The software installation request process follows a structured workflow within ServiceNow:

Step 1 – Request Submission

The employee accesses the **Service Catalog** and selects the **Software Installation Request** catalog item.

Step 2 – Request Details

The employee completes the request form by providing the required information, including:

- Software Name
- Software Version
- Business Justification
- Installation Device
- License Type
- Required Installation Date

After entering the details, the employee submits the request.

Step 3 – Request Creation

ServiceNow automatically creates:

- A **Request (sc_request)** record
- A linked **Requested Item (sc_req_item)** containing all submitted information

These records are used to track the request throughout its lifecycle.

Step 4 – Manager Approval

The approval workflow is triggered automatically.

An **Approval (sysapproval_approver)** record is created and assigned to the employee's reporting manager.

The manager reviews the request through the **My Approvals** workspace and either approves or rejects it.

Step 5 – Approved Request

If the manager approves the request:

- The Requested Item status is updated.
- A **Catalog Task (sc_task)** is automatically generated.
- The task is assigned to the **Network/Software Support Team**.
- The employee receives an email notification confirming the approval.

Step 6 – Rejected Request

If the manager rejects the request:

- The Requested Item status changes to **Rejected**.
- The employee receives an email containing the rejection notification and comments provided by the manager.

Step 7 – License Shortage Handling

If the requested software cannot be installed due to unavailable licenses:

- The request is moved to the **On Hold** state.
- A **Business Rule** automatically creates an **Incident**.
- The incident is assigned to the **Software Support Team** for investigation and further action.

This ensures that license-related issues are tracked and resolved without manual intervention.

3. User Stories

User Story ID	User Story	Priority
US-01	As an employee, I want to submit a software installation request through a simple Service Catalog form so that I can request software without sending emails to the IT team.	High
US-02	As an employee, I want to track the status of my software request so that I know whether it is pending approval, in progress, or completed.	High
US-03	As a manager, I want to receive software requests in the My Approvals workspace so that I can review, approve, or reject them efficiently.	High
US-04	As a manager, I want to view the employee's business	Medium
US-04	justification before making an approval decision so that I can make an informed choice.	Medium
US-05	As a member of the IT/Network Support team, I want approved software requests to automatically generate fulfillment tasks so that installation work can be assigned and tracked efficiently.	High
US-06	As a member of the IT/Network Support team, I want an incident to be created automatically whenever a software request cannot be fulfilled due to license shortages so that the issue can be resolved without manual follow-up.	Medium
US-07	As an employee, I want to receive email notifications whenever my request is approved, rejected, or completed so that I remain informed without checking the system repeatedly.	Medium

Summary

The proposed ServiceNow workflow provides a structured and automated process for handling software installation requests. It simplifies request submission for employees, enables managers to make timely approval decisions, automatically assigns approved requests to the IT support team, and ensures license shortages are handled through incident creation. This approach improves efficiency, transparency, accountability, and overall service quality across the organization.